

Technical Report on Constraining γ with 190 MeV/u Polarized-Deuteron Breakup

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June 11, 2026

Data Scope

- Local repository: `/home/tian/workspace/dpol/smsimulator5.5`
- NEBULA-Plus reconstruction sample: `/home/tian/workspace/sshDir/spana03/Dpol_smsimulator/data/reconstruction/nebulaplus_nn_joint_20260609_012854`
- Folding diagnostic tables: `checks/observables/folding_diagnostics/`
- Figures used in this report: `docs/reports/gamma_constraint_20260611/figures/`

Data provenance Three-stage pipeline (all figures in this report use the same sample):

- (1) QMD simulation: d+Sn 190 MeV/u, $\gamma = \{0.5, 0.6, 0.7, 0.8\}$, targets Sn112+Sn124, y-pol (phi_random) + z-pol (b_discrete).
- (2) Breakup extraction: `breakup_nn_targetframe_20260607_180428` (2026-06-07 18:04).
- (3) NEBULA-Plus + PDC NN joint reconstruction: `nebulaplus_nn_joint_20260609_012854` (2026-06-09 01:28).

All figure scripts (`make_cut_diagnostics_figures.py`, `plot_nebulaplus_reco.py`) read stage (3) directly via uproot — 176 *_reco.root files (`recoTree`), filtered by `truth_has_proton & truth_has_neutron & nn_status==0`. Sn124 event counts: y-pol 1.68×10^6 (16 files), z-pol 3.75×10^4 (160 files).

Executive Summary

The current simulation and reconstruction sample support the following statements.

1. For Sn124, the folded visible-window observable with the tight selection and $|p_{x,n}| < 60$ MeV/c still keeps a clear dependence on γ . In y-pol, the folded reconstructed R changes from about 12.48 to 4.99 when γ changes from 0.5 to 0.8. In z-pol, it changes from about 6.38 to 0.56.
2. For a 15 mm diameter Sn124 target made from the 1.2 g inventory, a beam intensity of 1.0×10^7 pps, and 16 h of beam time, the current planning anchor gives enough statistics even after applying a conservative usable-event survival factor from the full simulation: about 2.75×10^5 usable R events for y-pol and 1.88×10^6 for z-pol.
3. The hardest adjacent model separation in the current grid is y-pol $g070 \rightarrow g080$. A purely statistical 3σ separation needs only about 2.8×10^3 usable R events. Statistics are therefore not the leading limitation.
4. The main risk is systematic, not statistical. The new diagnostics show the detector folding explicitly: event-selection survival, neutron p_x - p_y efficiency, $p_{x,n}$ and Δp_x response matrices, reaction-plane migration, sign migration, and the final R ladder. Truth-level R and reconstructed R are not the same quantity; the robust strategy is to fold every theory point through the same Geant4 and reconstruction chain before comparing it with data.
5. The PDC geometry is treated as the final effective 69 deg configuration. The NEBULA-Plus macro executes the base geometry first, then explicitly sets the PDC angle to 69 deg and calls `Update`. The 65 deg message in the logs is an earlier setting that is overwritten.

1 Observable and Physics Logic

The physics goal is to use the proton–neutron isovector force difference near the target to detect a bias in the final breakup relative momentum. For each event, the transverse reaction-plane direction is defined by

$$\phi = \text{atan2}(p_{y,p} + p_{y,n}, p_{x,p} + p_{x,n}).$$

After rotating the proton and neutron momenta into this event plane, the working observable is

$$\Delta p_x = p_{x,p}^{\text{rot}} - p_{x,n}^{\text{rot}}, \quad R = \frac{N(\Delta p_x > 0)}{N(\Delta p_x < 0)}.$$

The experiment does not observe the truth distribution directly. It observes a folded distribution:

$$N_{\text{reco},j} = \sum_i K_{ji} N_{\text{true},i} + b_j.$$

Here K_{ji} includes NEBULA/NEBULA-Plus acceptance, resolution, reconstruction efficiency, PDC proton reconstruction, reaction-plane migration, and analysis cuts. In general,

$$R_{\text{reco}} \neq R_{\text{truth}}.$$

This report therefore does not interpret reconstructed R as a direct estimator of truth-level R . It instead tests whether $R(\gamma)$ remains separable under a common folded analysis definition.

2 Most Convincing Current Gamma Sensitivity

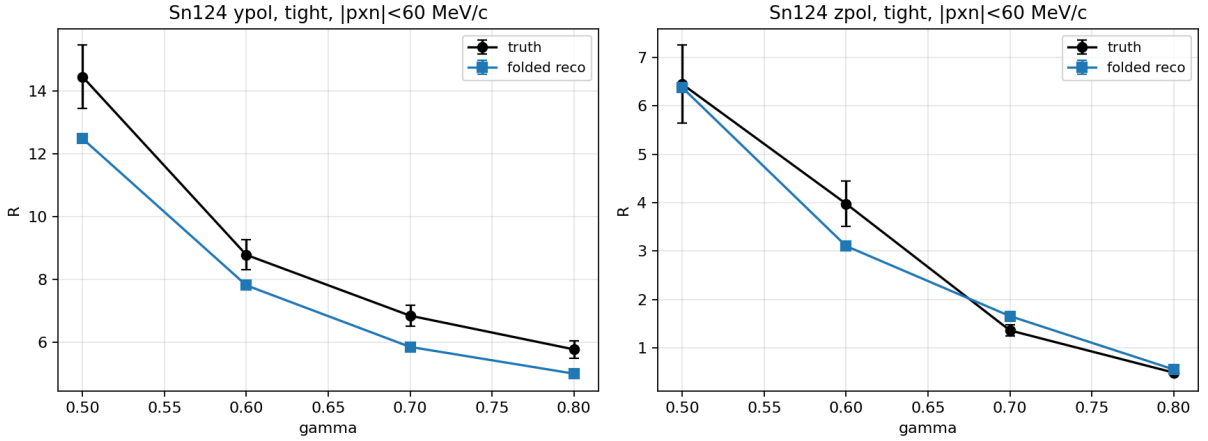


Figure 1: Truth and folded reconstructed $R(\gamma)$ for Sn124 with the tight selection and $|p_{x,n}| < 60$ MeV/c.

For Sn124 with the tight selection and $|p_{x,n}| < 60$ MeV/c, the folded reconstructed reaction-plane observable is:

polarization	γ	N_+	N_-	R_{folded}	$\sigma(R)_{\text{sim}}$	$\sigma(R)_{16\text{h}}$
y-pol	0.5	2134	171	12.48	0.99	0.091
y-pol	0.6	2147	275	7.81	0.50	0.047
y-pol	0.7	2093	358	5.85	0.33	0.032
y-pol	0.8	1948	390	4.99	0.28	0.026
z-pol	0.5	306	48	6.38	0.99	0.014
z-pol	0.6	208	67	3.10	0.44	0.005
z-pol	0.7	214	129	1.66	0.18	0.0025
z-pol	0.8	171	307	0.56	0.053	0.0008

N_+/N_- are the counts of $\Delta p_x^{\text{reco}} > 0 / < 0$ events, $\sigma(R)_{\text{sim}}$ is the statistical error of the current MC sample, and $\sigma(R)_{16\text{h}}$ is the error extrapolated to a 16 h run (formula in the Statistics and Beam Time section).

The important point is not that truth and reconstruction are identical. The important point is that the folded reconstructed curve remains monotonic in the current Sn124 sample, and its separation is much larger than the statistical uncertainty expected for a 16 h run.

3 Statistics and Beam Time

The current beamtime planning anchor is:

- beam energy: 190 MeV/u
 - beam intensity: 1.0×10^7 pps
 - SRC harmonic number: 6
 - bunch spacing: 33.62 ns
 - planning breakup cross section: 550 mb
 - planning detected-coincidence efficiency: 4.65%
- For the 15 mm Sn124 target option under the 1.2 g inventory constraint:
- equivalent thickness: 0.929 mm
 - detected coincidence rate: 843 Hz
 - detected coincidences in 16 h: 4.86×10^7

3.1 QMD sample size and pass rates

The QMD sample used for the current plots (Sn124, summed over the four γ points) is highly unbalanced between the two polarizations:

pol	N_{raw} (QMD)	N_{tight}	N_{reco} (px60)	raw→tight	raw→reco
y-pol	1.68×10^6	4.36×10^4	9.52×10^3	2.59%	0.566%
z-pol	3.75×10^4	6.04×10^3	1.45×10^3	16.1%	3.87%

y-pol generated 1.68×10^6 QMD events while z-pol generated only 3.75×10^4 (a factor of 45 fewer). z-pol has a higher pass rate (16.1% vs 2.59%), but the raw sample is so small that the current z-pol plots carry large MC statistical errors ($\sigma(R)_{\text{sim}} \sim 0.05\text{--}0.99$, see Table 1). N_{reco} is the number of usable R events actually used in the current tight-px60 plots.

Comparing the 16 h experimental yield directly with the current MC plot statistics:

pol	useful fraction	N_{sim} (current plot)	16 h usable	16 h / N_{sim}	worst 3σ N_{req}
y-pol	0.00566	9.52×10^3	2.75×10^5	$\sim 29 \times$	2.77×10^3
z-pol	0.03869	1.45×10^3	1.88×10^6	$\sim 1.3 \times 10^3 \times$	1.31×10^2

A 16 h run gives $\sim 29\times$ the current y-pol plot statistics and $\sim 1300\times$ the current z-pol statistics. The useful fraction $f_{\text{useful}} = N_{\text{reco}}/N_{\text{raw}}$ converts detected coincidences to usable R events: $N_{\text{usable}} = 4.86 \times 10^7 \times f_{\text{useful}}$. The $\sim 1300\times$ z-pol gain is especially important: the current z-pol MC is severely under-sampled (1.45×10^3 usable events), so the experiment will substantially improve its precision.

3.2 Statistical-error derivation

$R = N_+/N_-$ is a ratio estimator. Under a multinomial model on $N = N_+ + N_-$ usable events, the delta method gives

$$\sigma(R) = (1 + R)\sqrt{R/N}.$$

The number of events needed for a 3σ separation of two adjacent γ points (using the midpoint $\bar{R} = \frac{1}{2}(R_1 + R_2)$ for σ) is

$$N_{\text{req}} = \left[z_\alpha (1 + \bar{R})\sqrt{\bar{R}} / |\Delta R| \right]^2, \quad z_\alpha = 3.$$

The per-interval result for the tight px60 folded reco observable is:

pol	interval	R_{left}	R_{right}	$ \Delta R $	$N_{\text{req}}(3\sigma)$
y-pol	g050→g060	12.48	7.81	4.67	519
y-pol	g060→g070	7.81	5.85	1.96	979
y-pol	g070→g080	5.85	4.99	0.85	2.77×10^3
z-pol	g050→g060	6.38	3.10	3.27	131
z-pol	g060→g070	3.10	1.66	1.45	117
z-pol	g070→g080	1.66	0.56	1.10	36

The hardest separation is y-pol g070→g080 ($\bar{R} = 5.42$, $|\Delta R| = 0.85$, needs 2.77×10^3 events); for z-pol it is g050→g060 ($\bar{R} = 4.74$, $|\Delta R| = 3.27$, needs 131). The usable-event extrapolation chain is

$$N_{\text{usable}} = \text{rate} \times T \times f_{\text{useful}}, \quad f_{\text{useful}} = N_{\text{reco_plane}}(\text{tight px60})/N_{\text{raw}},$$

so y-pol $f_{\text{useful}} = 0.00566$ gives $N_{\text{usable}} = 4.86 \times 10^7 \times 0.00566 \approx 2.75 \times 10^5$ and z-pol $f_{\text{useful}} = 0.0387$ gives 1.88×10^6 . The usable/required ratio is $\approx 99\times$ for y-pol (about two orders of magnitude) and $\approx 1.4 \times 10^4\times$ for z-pol (about four orders of magnitude).

3.3 Statistical vs systematic comparison

Putting the three scales side by side for y-pol tight px60:

quantity	typical value
$\sigma_{\text{stat}}(R)$ at 16 h (per γ point)	0.026 – 0.091
$\Delta R_{\text{detector}}$ (truth→reco, per γ point)	1 – 2
$\Delta R_{\text{between } \gamma}$ (hardest interval g070→g080)	0.85

The 16 h statistical error ($\sim 0.03\text{--}0.09$) is one to two orders of magnitude below the detector-folding shift ($\sim 1\text{--}2$), and about an order of magnitude below the smallest γ gap (0.85). The conclusion that “statistics is not the bottleneck, systematics dominate” is therefore quantitatively supported.

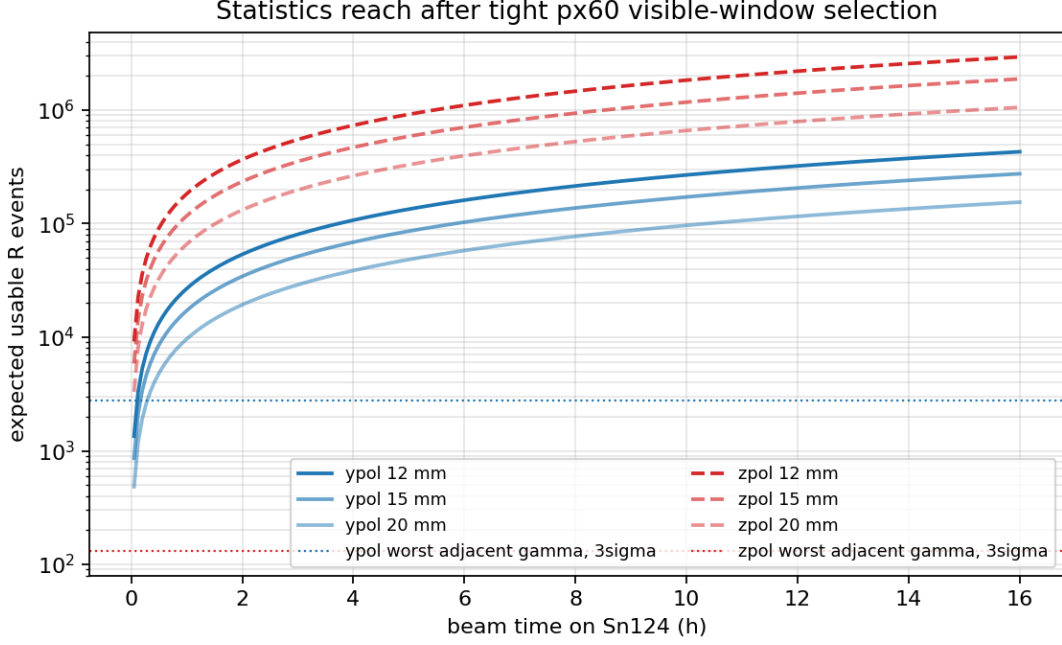


Figure 2: Usable R -event reach after the tight px60 visible-window selection for the Sn124 inventory target options.

Even with the 20 mm disk option, still using the same 1.2 g Sn inventory, the expected y-pol usable R events after 16 h remain above 10^5 . The experiment is statistically capable of constraining γ in this folded-observable strategy. The final uncertainty budget will be controlled by detector response, neutron efficiency, and model-folding systematics.

For pile-up, the older 3 mm reference thickness gives a detected-coincidence mean of about 9.16×10^{-5} per bucket. The 15 mm inventory option has a lower rate, so single-bucket pile-up is not the main limitation. The electronics integration window still needs to be checked for possible multi-bucket pile-up.

4 Recommended Gamma-Constraint Workflow

The recommended analysis path is:

1. For each QMD γ point, run the same Geant4 geometry, NEBULA-Plus detector response, PDC NN proton reconstruction, and neutron reconstruction.
2. Define the same folded observable for data and model: the tight event selection plus $|p_{x,n}^{\text{reco}}| < 60 \text{ MeV}/c$, with Sn124 as the primary target.
3. Construct the experimental observable

$$R_{\text{exp}} = \frac{N(\Delta p_x^{\text{reco}} > 0)}{N(\Delta p_x^{\text{reco}} < 0)}.$$

4. Construct the folded model prediction $R_{\text{model,folded}}(\gamma)$.
5. Use a likelihood or χ^2 :

$$\chi^2(\gamma) = \frac{[R_{\text{exp}} - R_{\text{model,folded}}(\gamma)]^2}{\sigma_{\text{stat}}^2 + \sigma_{\text{detector}}^2 + \sigma_{\text{model}}^2}.$$

6. Combine y-pol/z-pol and Sn112/Sn124 only after each channel has its own closure test and systematic uncertainty estimate. Do not combine channels using statistical errors alone.

This route avoids an unstable unfolding problem. An unfolded truth-level R should be treated as an additional result only if closure tests demonstrate that the response matrix is sufficiently constrained and that near-zero-efficiency regions are not driving the result.

5 Detector Folding Diagnostics

The detector question is whether the reconstructed observable is still a controlled, calibratable folded observable. The current diagnostics separate the folding into event selection, neutron visible-window efficiency, reconstruction response, reaction-plane migration, and the final R ladder. The available output does not yet contain an independent layer-by-layer proton detector efficiency table; the proton/reaction-plane contribution is therefore evaluated through the PDC-reconstructed event plane, the response matrices, and the `reco_truth_plane` to `reco_plane` change in the R ladder.

5.1 Event selection and visible-window survival

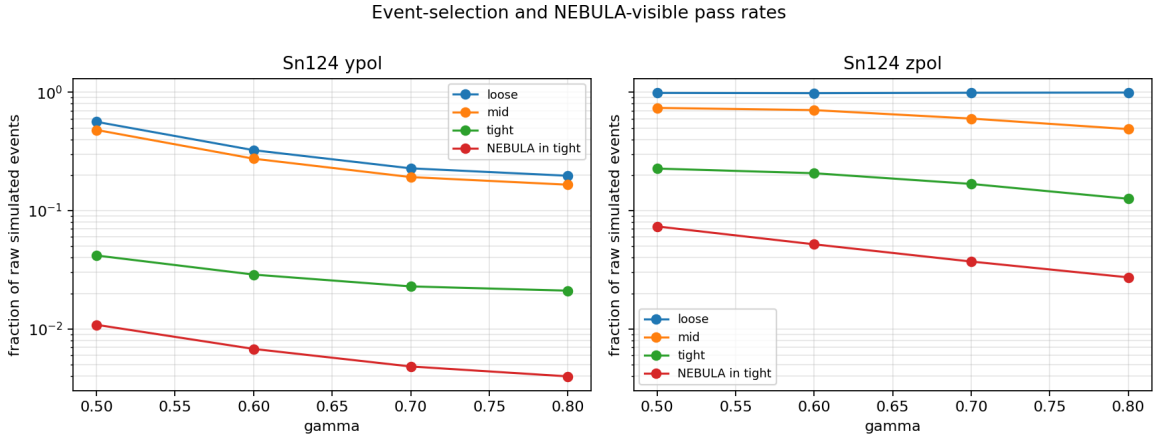


Figure 3: Sn124 selection pass rates versus γ . The final NEBULA-visible tight sample is the small fraction that can enter the folded R observable.

For Sn124, the pooled y-pol sample has raw-to-tight and raw-to-NEBULA-visible fractions of about 2.59% and 0.571%. The z-pol sample has larger fractions, about 16.1% and 3.93%, because the generated and selected phase spaces are different. These numbers are used only as the simulation-derived useful-event survival factors; they are not a replacement for the experimental coincidence-efficiency calibration.

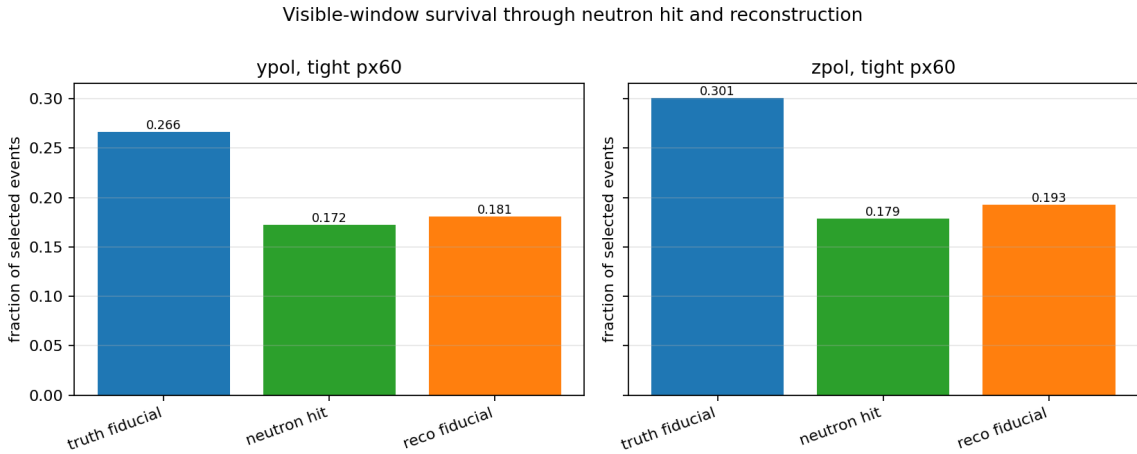


Figure 4: Survival inside the tight px60 diagnostic sample. The truth fiducial fraction is set by the visible-window definition; the next steps show neutron hit and reconstructed-fiducial survival.

For pooled tight px60 events, y-pol keeps 0.266 of selected events in the truth fiducial window,

0.172 after neutron hit, and 0.181 after reconstructed-fiducial selection. The corresponding z-pol fractions are 0.301, 0.179, and 0.193. The reconstructed-fiducial count can be slightly above the hit-truth-fiducial count because the fiducial condition is applied after reconstruction and can migrate events across the boundary.

5.2 Neutron efficiency in px-py and Delta-px phase space

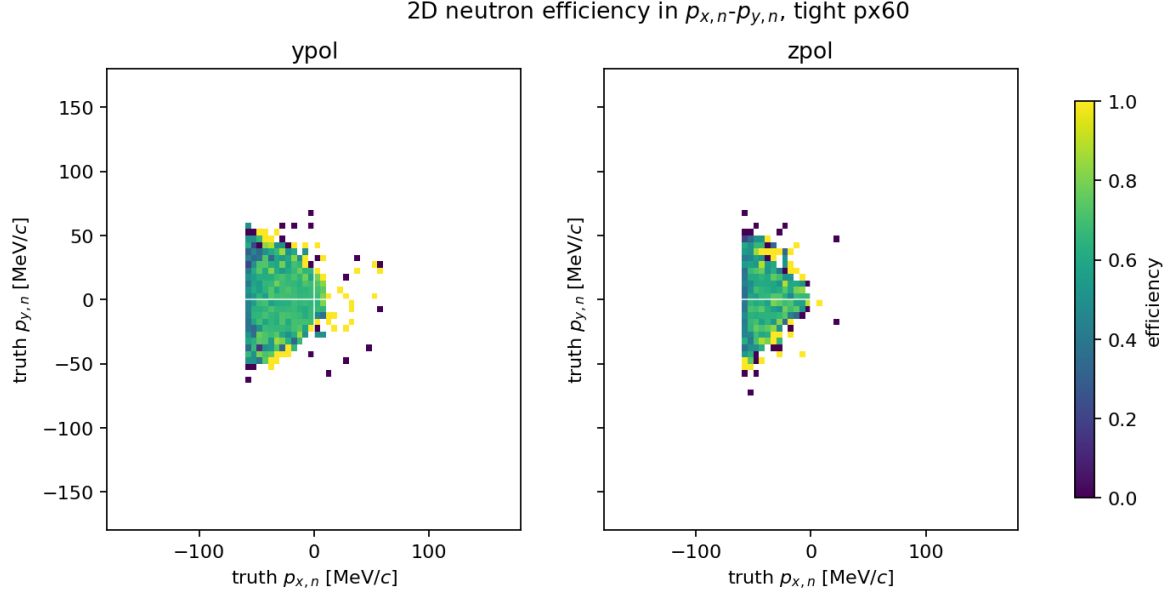


Figure 5: Two-dimensional neutron efficiency in truth $p_{x,n}$ versus truth $p_{y,n}$ for tight px60. Empty white regions are outside the populated/usable diagnostic phase space.

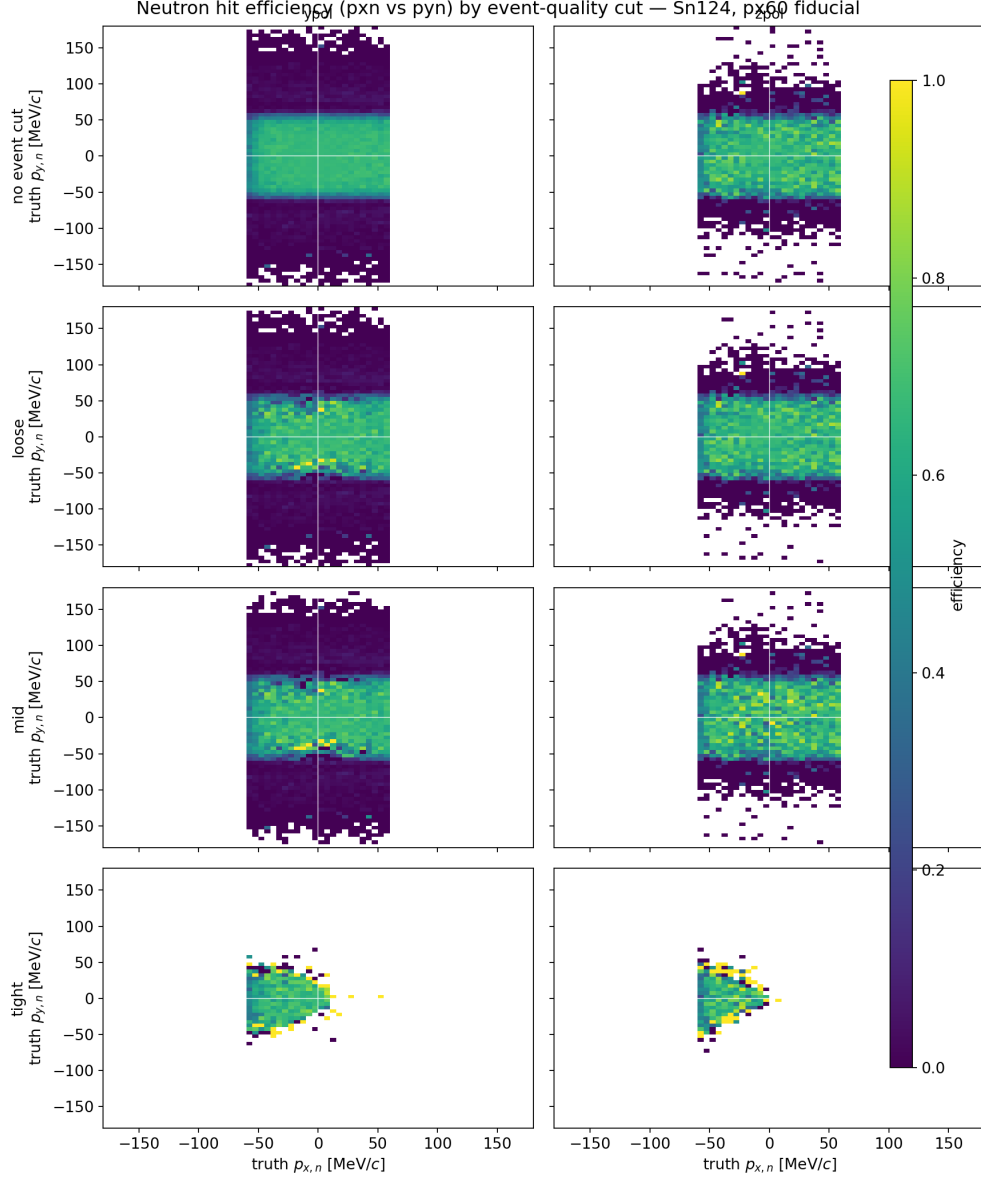


Figure 6: The same $p_{x,n}$ - $p_{y,n}$ efficiency map, one row per event-quality cut (none/loose/mid/tight), columns = y-pol/z-pol (all px60 truth fiducial). From none to tight the high-efficiency region keeps essentially the same shape: the cut changes which phase-space region is sampled, not the per-bin efficiency.

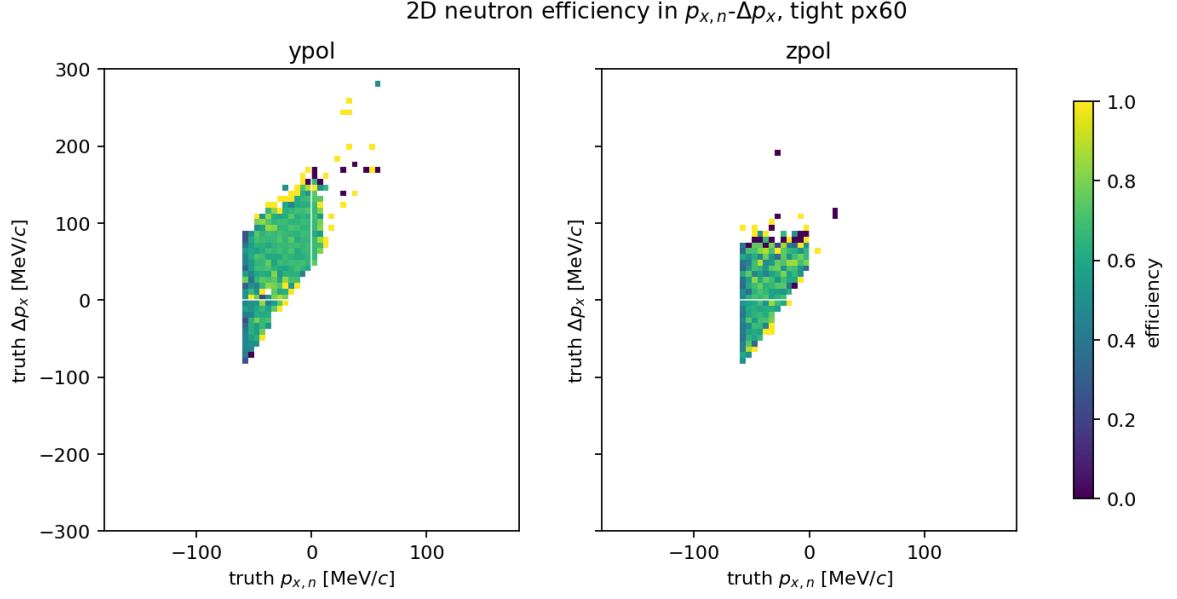


Figure 7: Two-dimensional neutron efficiency in truth $p_{x,n}$ versus truth Δp_x . This is the most direct efficiency map for the sign observable because it links neutron acceptance to the final numerator/denominator of R .

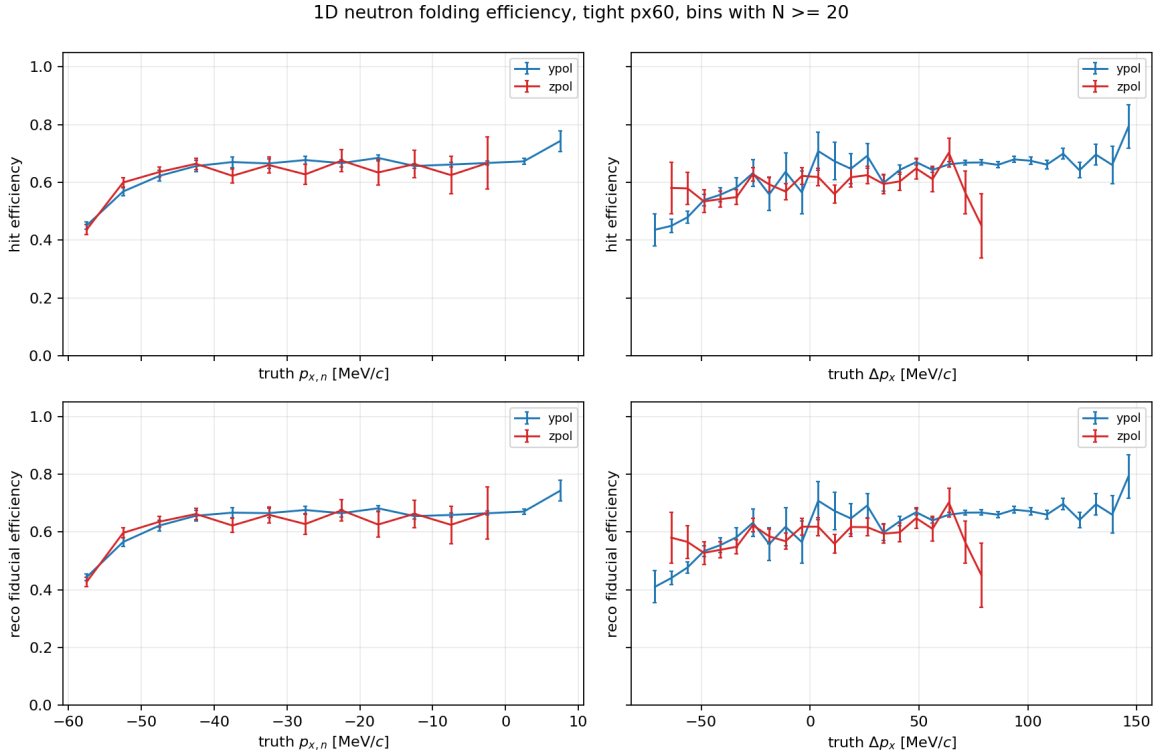


Figure 8: One-dimensional neutron hit and reconstructed-fiducial efficiencies for bins with at least 20 truth events. In the populated core, the efficiency is typically around 0.55–0.70 rather than flat at unity.

The visible sample is not a uniform slice of the breakup phase space. The populated tight px60 region lies mostly at negative $p_{x,n}$, and the efficiency varies across both $p_{x,n}$ and Δp_x . This is exactly why the proposal should not claim that the measured R directly equals the truth-level R .

The defensible claim is that the same efficiency map can be applied to every simulated γ point and tested with closure.

5.3 Sign-dependent neutron efficiency and the px60 choice

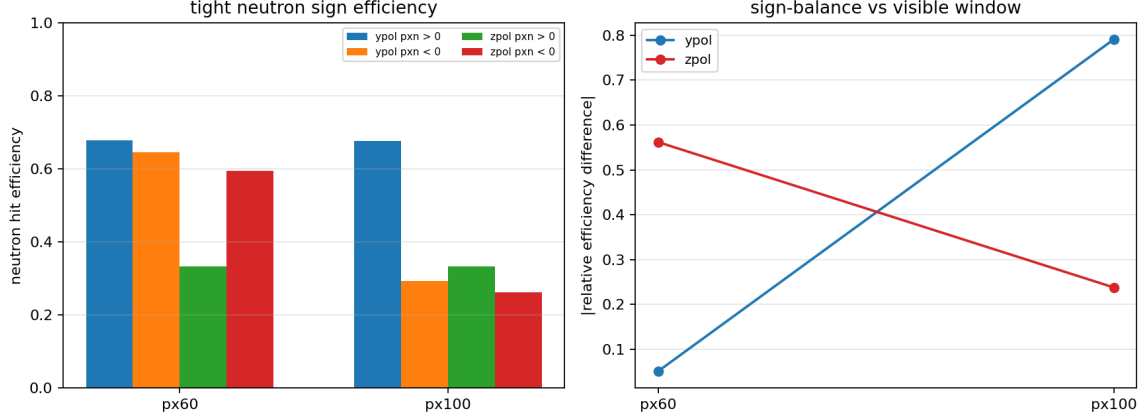


Figure 9: Neutron sign efficiency balance for tight selections. The y-pol sample gives the stable sign-balance comparison; z-pol tight has very small positive- $p_{x,n}$ statistics in this diagnostic.

For y-pol tight, the neutron hit efficiencies are:

fiducial	$\epsilon(p_{x,n} > 0)$	$\epsilon(p_{x,n} < 0)$	relative difference
px60	0.678	0.644	0.050
px100	0.677	0.293	0.791

The px100 selection retains a negative- $p_{x,n}$ branch that NEBULA sees very poorly, which strongly distorts R through a sign-dependent detector efficiency. The px60 selection does not remove all detector effects, but it reduces the largest y-pol sign-dependent efficiency imbalance to a range that can be modeled and tested with closure studies. The current z-pol tight sign-balance number should not be over-interpreted because the positive- $p_{x,n}$ branch contains only three events in the current MC diagnostic table.

5.4 Momentum and reaction-plane response

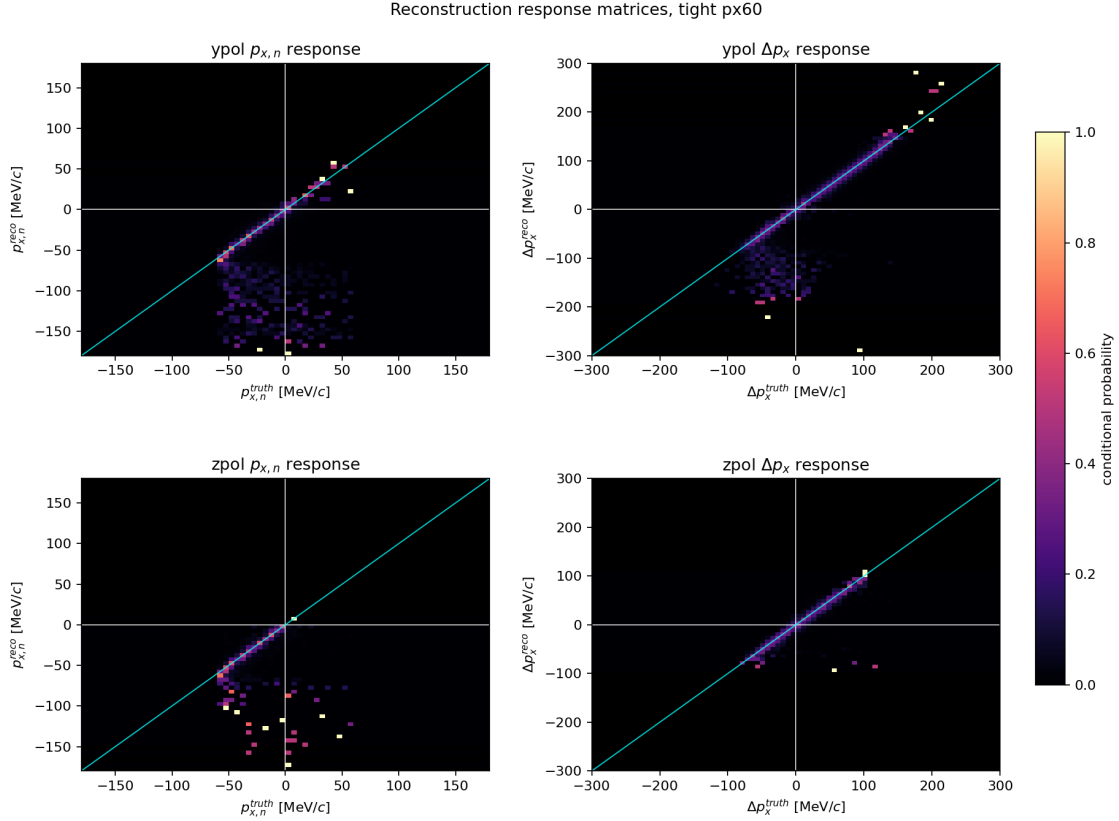


Figure 10: Conditional response matrices for reconstructed $p_{x,n}$ and reconstructed Δp_x in tight px60. The cyan line is the diagonal. Black regions are bins without usable diagnostic statistics.

The Δp_x response is concentrated near the diagonal for both polarizations in the core region. The $p_{x,n}$ response is narrower in the accepted branch but also shows low-probability tails and boundary effects. This pattern supports the same conclusion as the efficiency maps: reconstruction smearing is relevant and must be folded, but the first-order bias in R comes from which neutron kinematics enter the visible region.

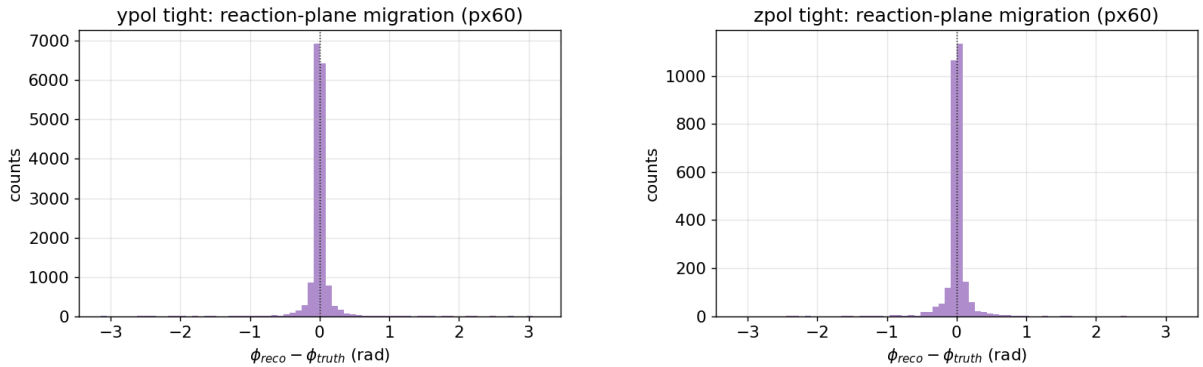


Figure 11: Reconstructed minus truth reaction-plane angle for y-pol and z-pol tight px60. This is the direct diagnostic of the proton/PDC-driven event-plane reconstruction used by the R observable.

5.5 Where the folding changes R



Figure 12: R ladder for y-pol and z-pol tight px60. The ladder separates truth, neutron-hit acceptance, truth-plane reconstruction, and fully reconstructed event-plane effects.

For pooled y-pol tight px60:

stage	N	N_{pos}	N_{neg}	R
truth	24323	21626	2697	8.02
hit_truth	15751	14349	1402	10.23
reco_truth_plane	16509	14476	2033	7.12
reco_plane	16509	14408	2101	6.86

The largest visible change happens before the final event-plane reconstruction: neutron hit acceptance and the reconstructed visible-window definition move the observable substantially. The smaller change from `reco_truth_plane` to `reco_plane` shows that the PDC/reaction-plane reconstruction is not the dominant R -bias term in the current diagnostic sample.

5.6 Sign migration is not the leading effect

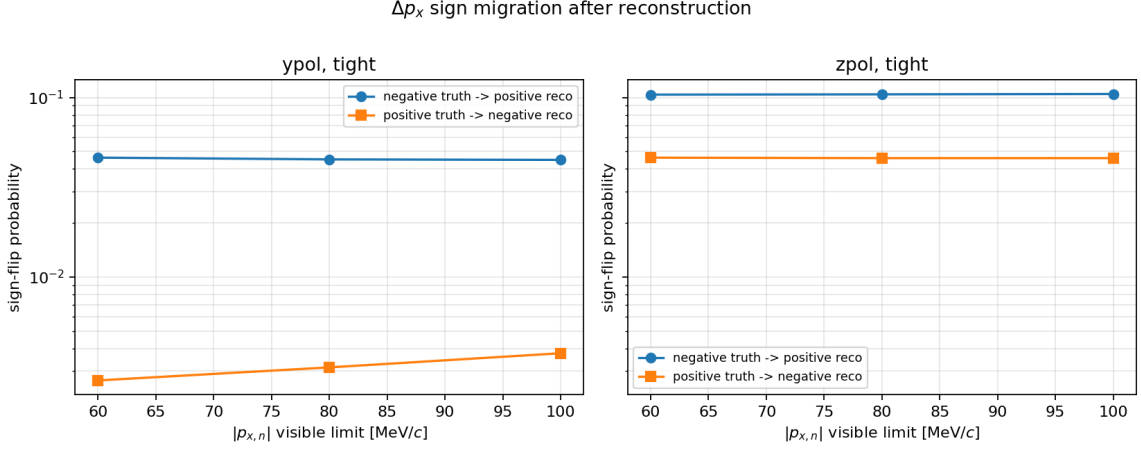


Figure 13: Δp_x sign-flip probability versus neutron visible-window size for tight selections. Sign migration is defined as the conditional probability $P(\text{reco sign} | \text{truth sign})$ with truth sign = $\text{sgn}(\Delta p_x^{\text{truth}})$ and reco sign = $\text{sgn}(\Delta p_x^{\text{reco}})$; the diagonal (truth→same sign) is the retention rate and the off-diagonal (truth→opposite sign) is the flip rate.

For tight px60, the sign-migration probabilities are:

polarization	truth sign	$P(\text{reco negative})$	$P(\text{reco positive})$
y-pol	negative	0.954	0.046
y-pol	positive	0.00265	0.997
z-pol	negative	0.896	0.104
z-pol	positive	0.046	0.954

Within the visible window, sign flipping of Δp_x is therefore not the main uncertainty. The leading issue is still which neutron kinematics enter the visible and reconstructable phase space.

6 Cut-System Diagnostics

The experimental observable depends on two sets of selections: the event-quality cut (**loose**/**mid**/**tight**) and the neutron visible window ($|p_{x,n}| < 60/80/100$ MeV/c). The figures below use event-level data read directly from ROOT (`make_cut_diagnostics_figures.py`, reusing the loader and cut definitions of `plot_nebulaplus_reco.py`) to show how these cuts change the observable and the phase-space coverage.

Cut definitions (truth-level momenta, see `compute_cut_masks`; Σ denotes p+n sum, $\phi_\Sigma = \text{atan2}(p_{y,\Sigma}, p_{x,\Sigma})$):

y-pol — **loose**: $|p_{y,p} - p_{y,n}| < 150$ and $|p_{T,\Sigma}| > 50$; **mid**: also $|p_{T,\Sigma}| < 200$; **tight**: also collinearity $(\pi - |\phi_\Sigma|) < 0.2$.

z-pol — **loose**: $(p_{z,p} + p_{z,n}) > 1150$ and $|p_{z,p} - p_{z,n}| < 150$; **mid**: also $|p_{x,\Sigma}| < 200$ and $|p_{T,\Sigma}| > 50$; **tight**: also $(\pi - |\phi_\Sigma|) < 0.5$.

6.1 $R(\gamma)$ across the 9 cut combinations

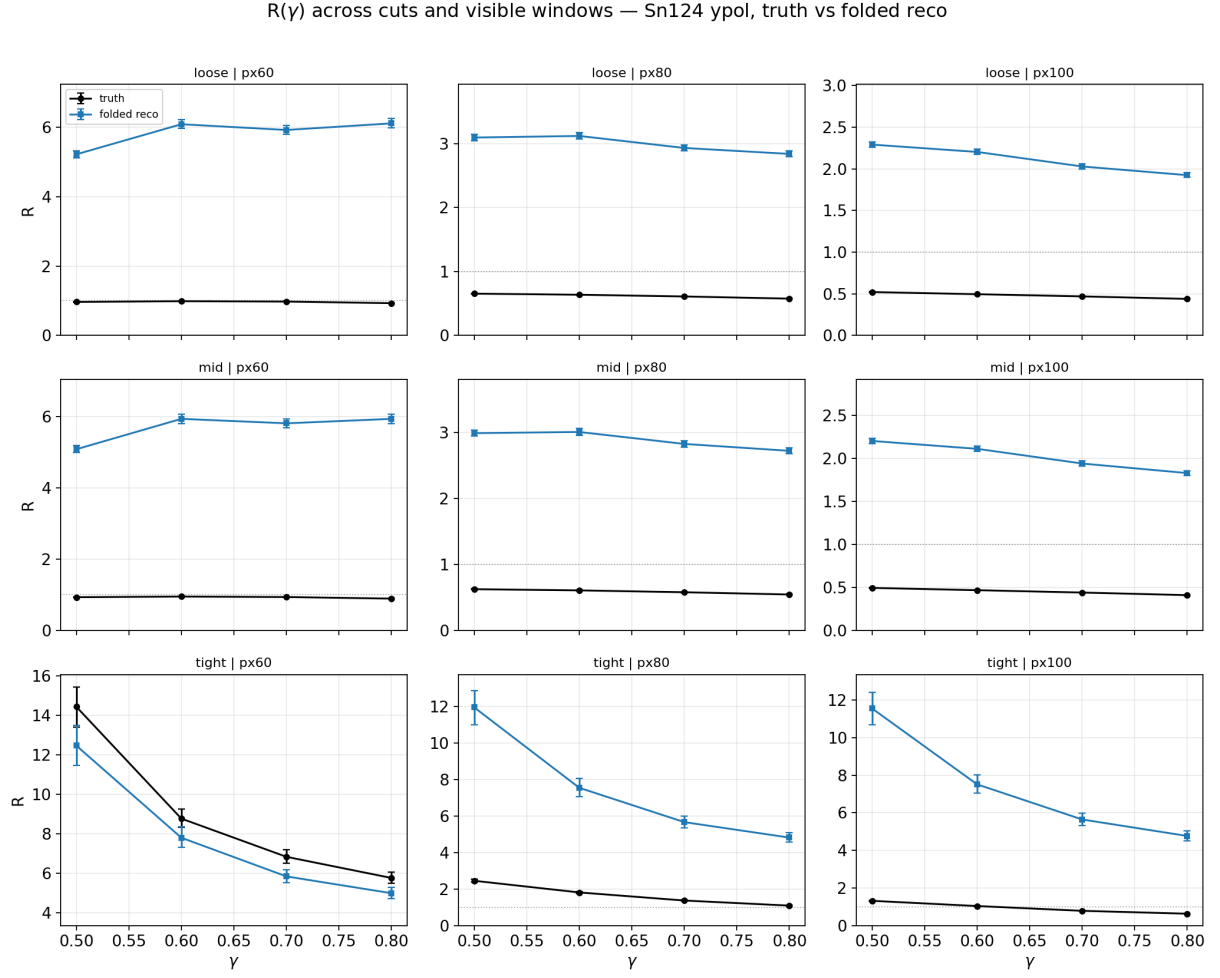


Figure 14: y -pol $R(\gamma)$ across the 9 cut combinations (rows = loose/mid/tight, columns = px60/px80/px100), Sn124. Black = truth-level R , blue = folded reco R , error bars from bootstrap.

$R(\gamma)$ across cuts and visible windows — Sn124 zpol, truth vs folded reco

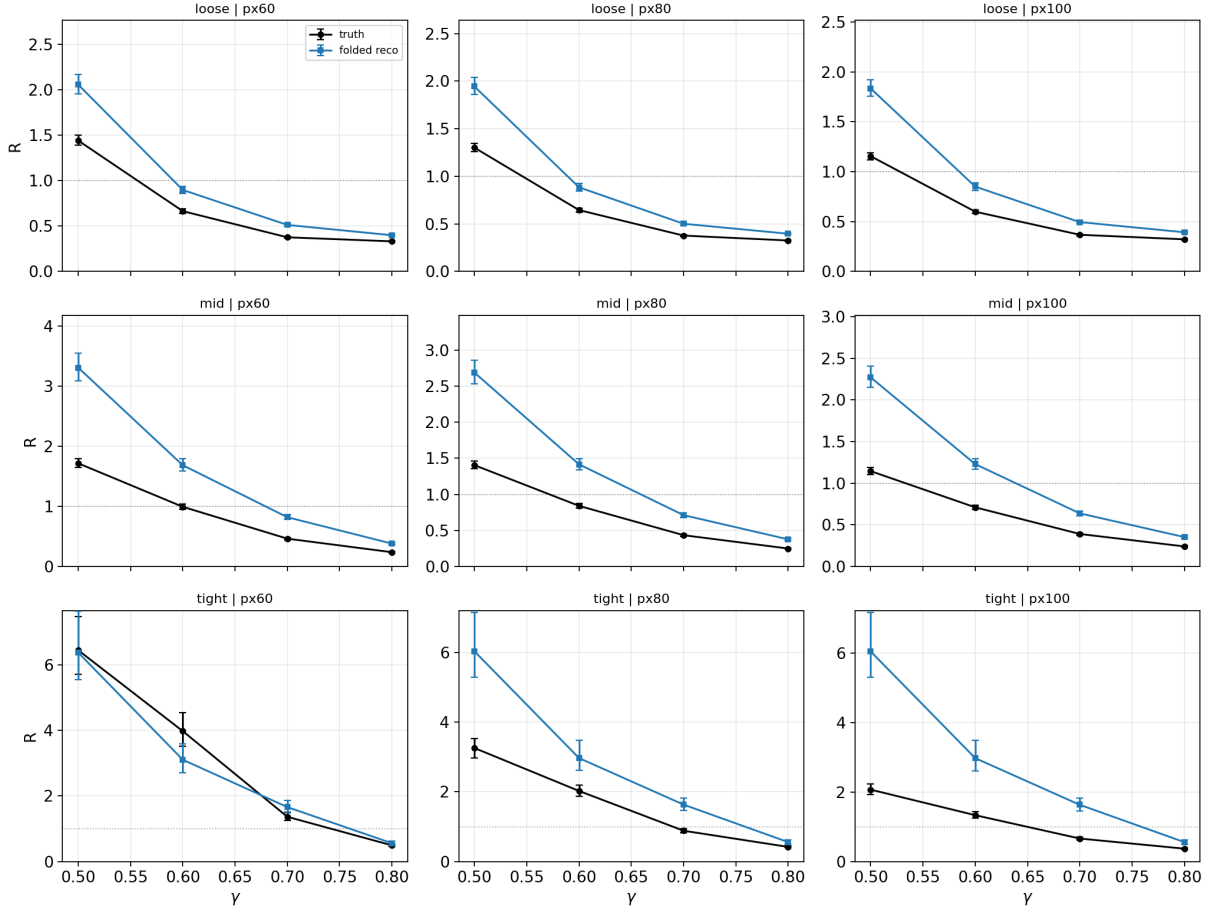


Figure 15: z-pol: the same 9-combination grid.

Key points: **loose**→**tight** makes y-pol R rise monotonically and z-pol R fall (the event selection increasingly favors one sign); px60→px100 increases R (a wider visible window keeps more of the negative- $p_{x,n}$ branch that NEBULA sees poorly); the truth/reco gap is smallest at tight px60. The γ sensitivity (slope) survives in all 9 combinations, but tight px60 gives the best truth/reco agreement.

6.2 Δp_x shape: truth vs reco

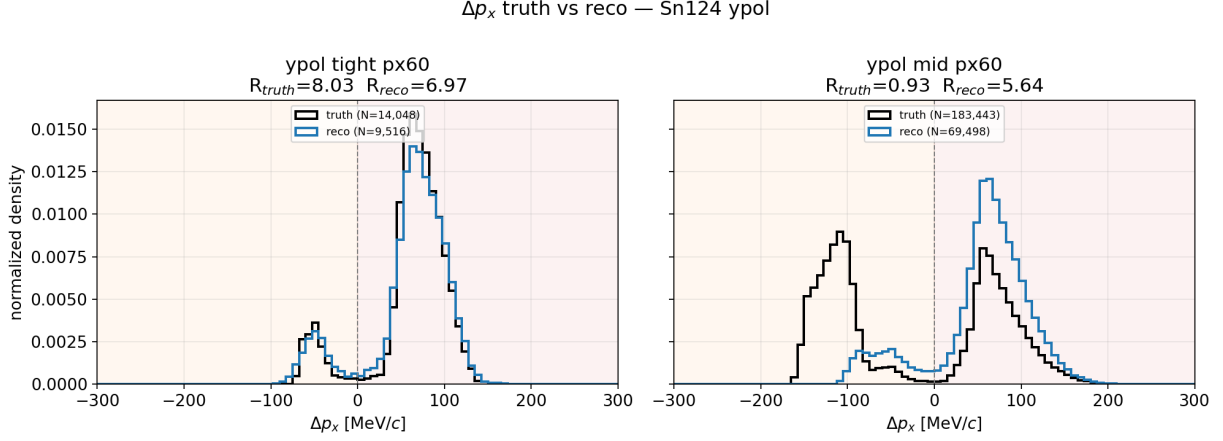


Figure 16: y-pol normalized Δp_x histograms. Black = truth (truth event plane + truth momenta), blue = reco (reco event plane + NN proton + reconstructed neutron). Red shading = $\Delta p_x > 0$ (R numerator), orange = $\Delta p_x < 0$ (R denominator). Tight px60 and mid px60 side by side.

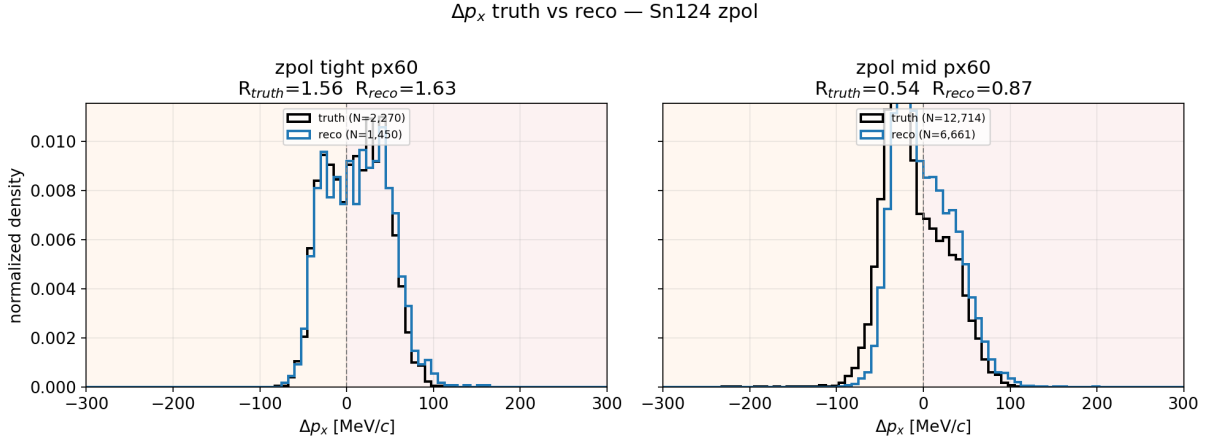


Figure 17: z-pol: the corresponding Δp_x truth vs reco overlay.

The reco peak position and width are close to truth; sign migration is mostly in the tails. The y-pol asymmetry (positive side much larger than negative) is visible in both truth and reco. Each panel title reports the corresponding R_{truth} and R_{reco} .

6.3 Phase-space coverage (truth level)

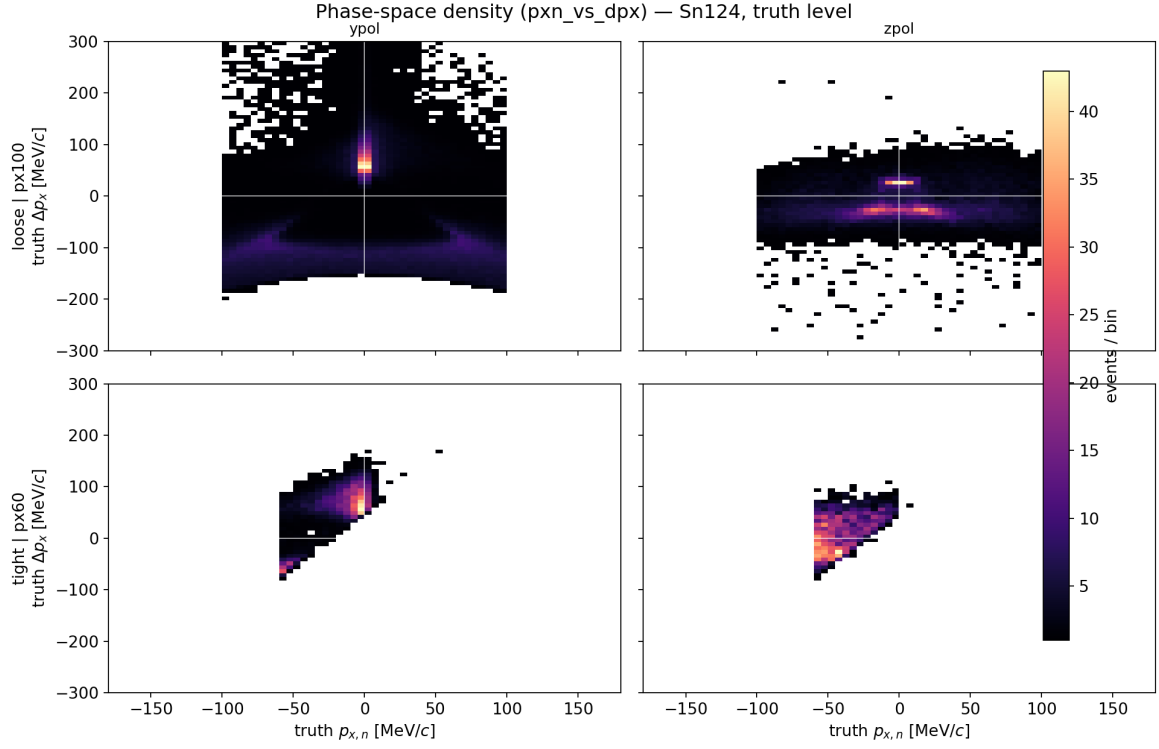


Figure 18: **Truth-level** phase-space density (event counts per bin, not efficiency). Top row = loose px100, bottom row = tight px60, columns = y-pol/z-pol. It answers “which region of truth phase space does this cut keep?”.

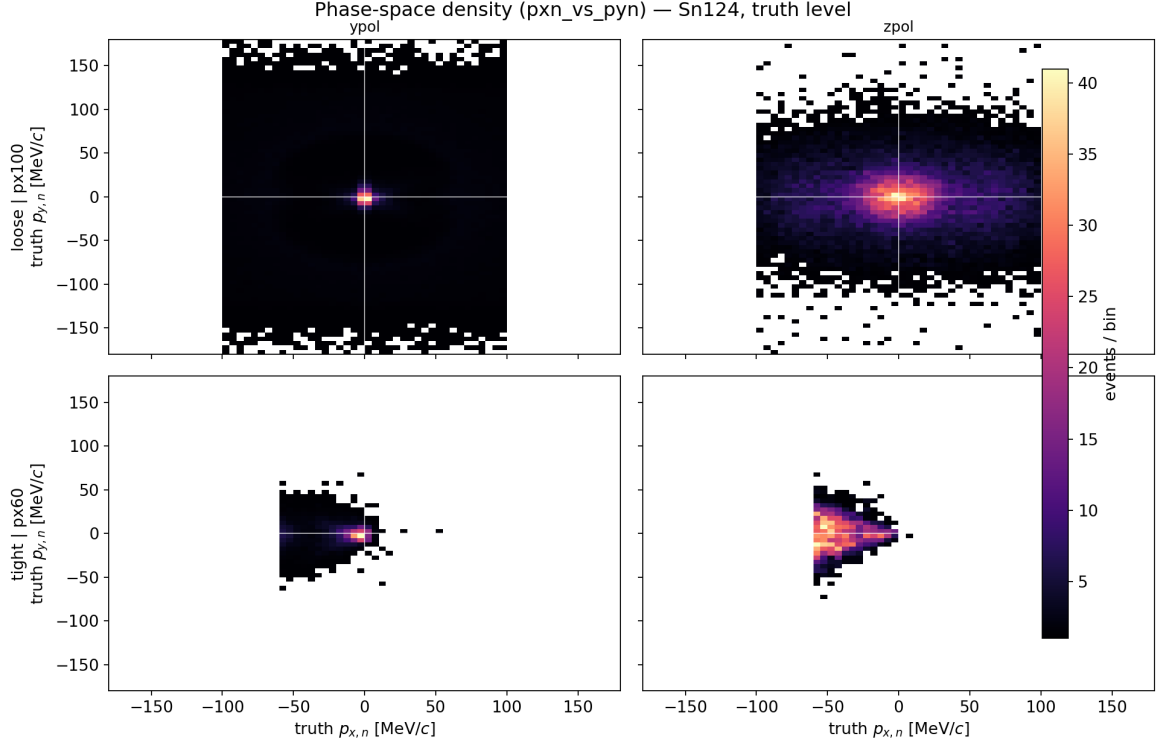


Figure 19: Same, truth $p_{x,n}$ vs $p_{y,n}$ neutron transverse plane.

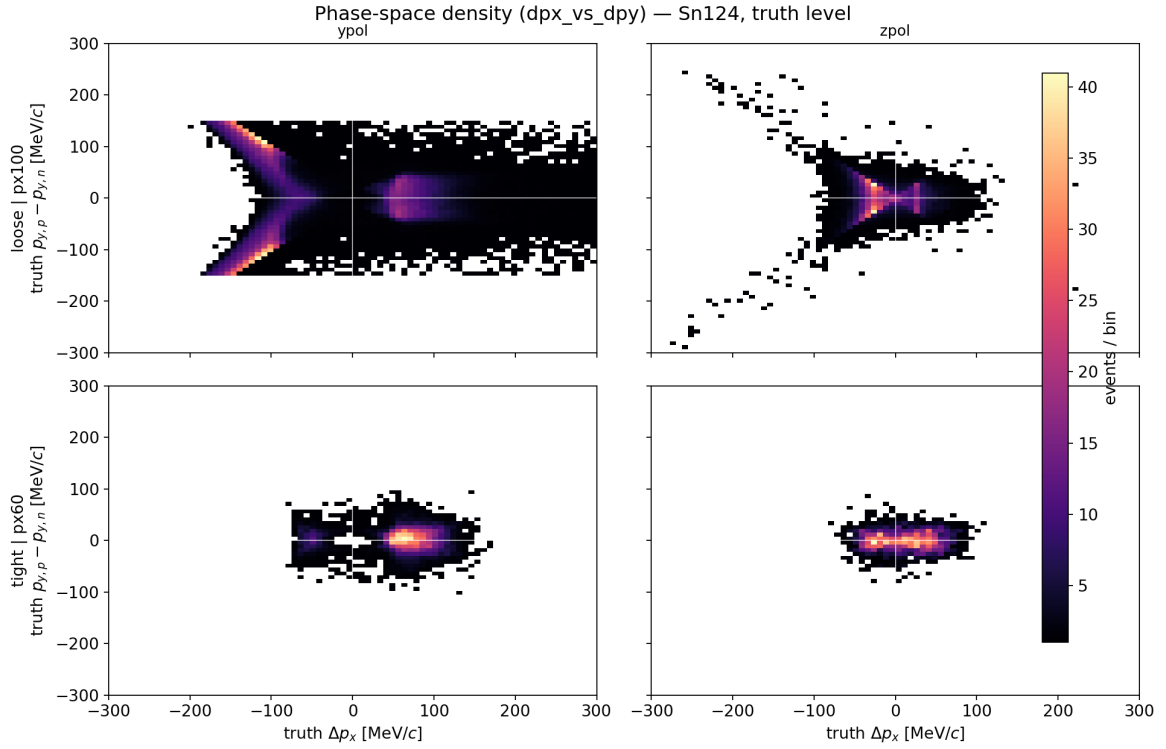


Figure 20: Same, truth Δp_x vs $(p_{y,p} - p_{y,n})$ momentum-asymmetry plane. The tight cut compresses events toward the reaction plane.

These three figures are **truth-level** phase-space densities (event counts, not efficiencies). Tight px60 carves out the narrow $|p_{x,n}| < 60$ strip and concentrates Δp_x on the positive side (y-pol); loose px100 covers a much broader region. Reco-level truth $\rightarrow$ reco migration is a separate question, covered by the response matrices in the Detector Folding Diagnostics section — this section addresses the geometric selection of the cut, not detector smearing.

6.4 Neutron efficiency across cuts

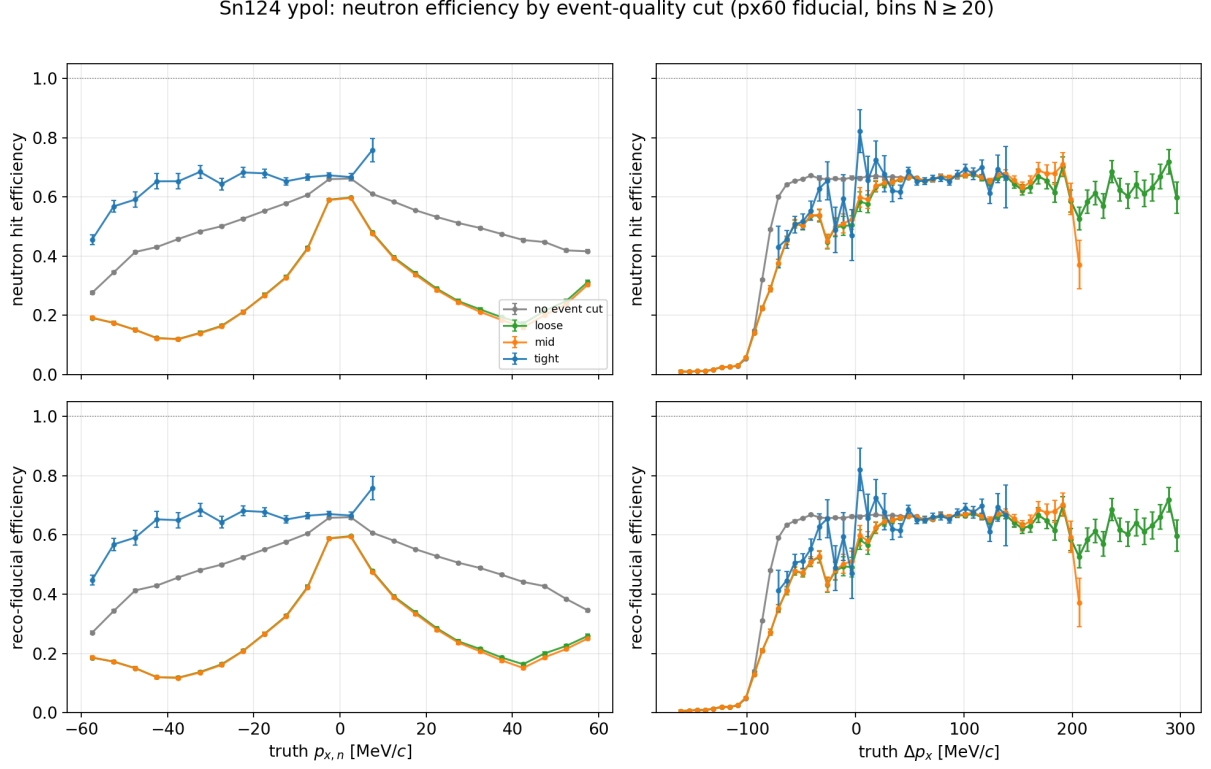


Figure 21: y-pol neutron efficiency versus event-quality cut (px60 truth fiducial). Four curves = none (no event cut) / loose / mid / tight. Top row = neutron hit efficiency, bottom row = reco-fiducial efficiency; left column vs truth $p_{x,n}$, right column vs truth Δp_x . Efficiency is $\epsilon(x) = N_{\text{hit}}(x\text{-bin})/N_{\text{fid}}(x\text{-bin})$; only bins with $N_{\text{fid}} \geq 20$ are drawn.

Sn124 zpol: neutron efficiency by event-quality cut (px60 fiducial, bins $N \geq 20$)

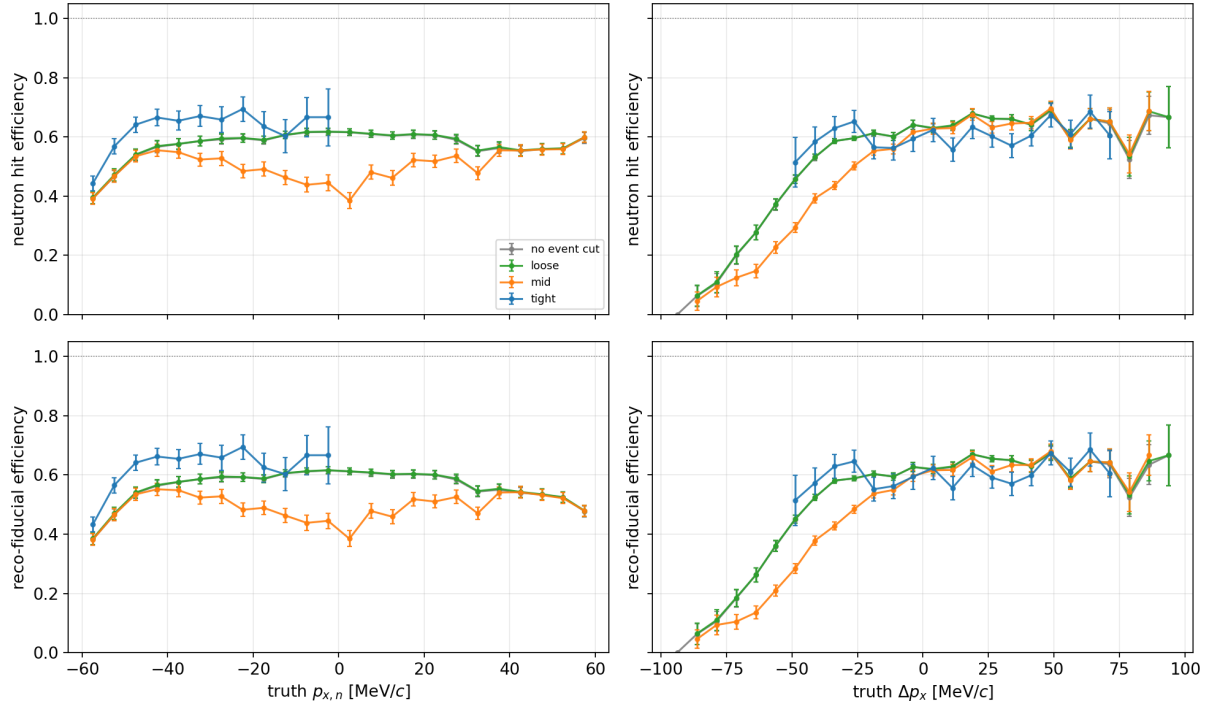


Figure 22: z-pol: the corresponding efficiency curves. The z-pol “none” curve has limited statistics in some bins.

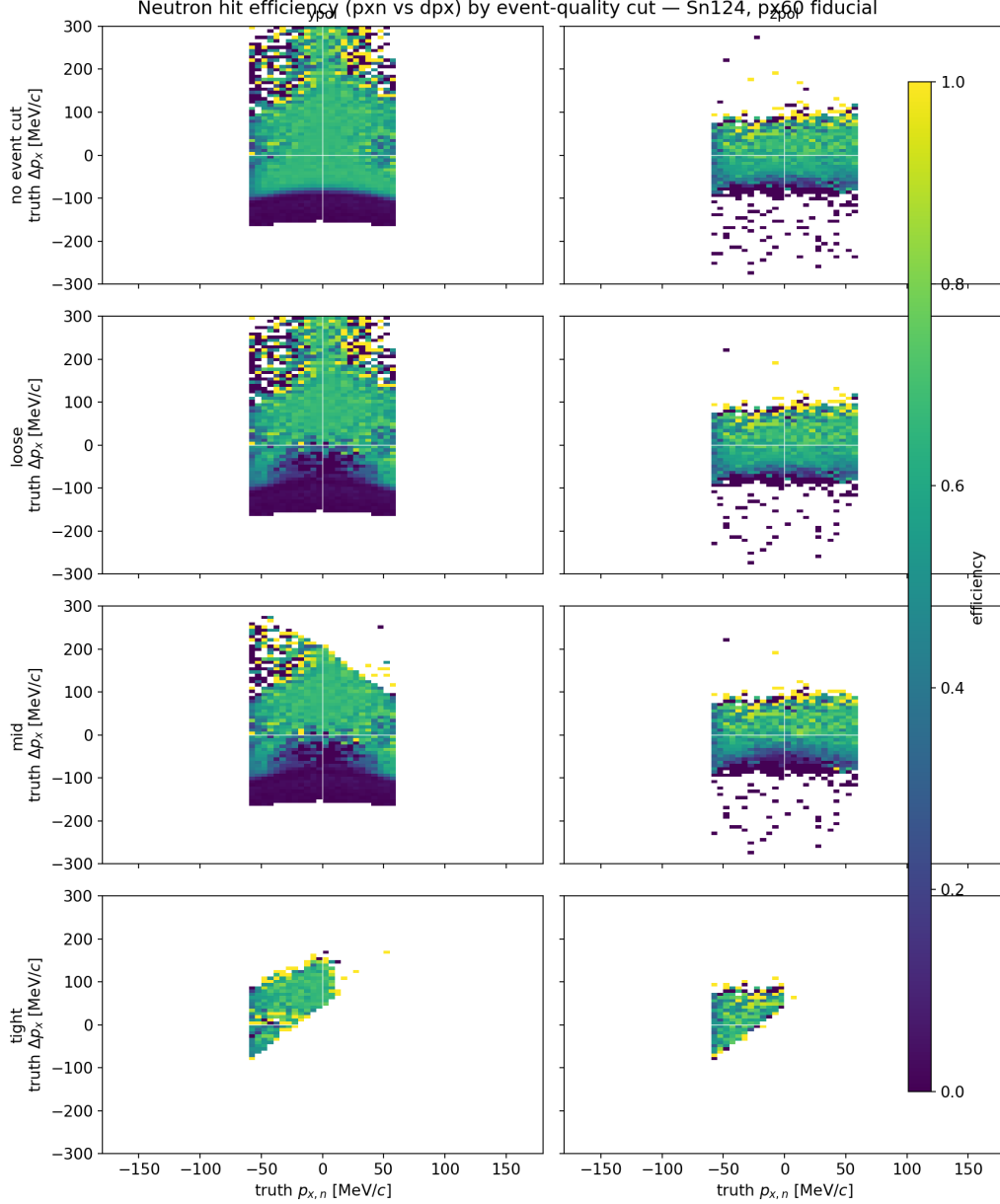


Figure 23: 2D neutron hit efficiency in $(p_{x,n}, \Delta p_x)$, one row per event-quality cut (none/loose/mid/tight), columns = y-pol/z-pol. All panels use the px60 truth fiducial.

Key point: from none to tight, the efficiency curve shape is largely unchanged (core $\epsilon \sim 0.55\text{--}0.70$), which means the event-quality cut mainly changes *which* phase-space region enters the sample (which $p_{x,n}/\Delta p_x$ bins have enough statistics), not the per-bin detection efficiency itself. The sign dependence of efficiency on $p_{x,n}$ (lower efficiency on the negative- $p_{x,n}$ side in y-pol) is present in both none and tight — it is an intrinsic NEBULA-geometry feature that the event-quality cut cannot remove; only narrowing the visible window (px60) excludes the most sign-dependent region.

6.5 Neutron efficiency without px fiducial (full range)

Sn124 ypol: neutron efficiency, NO px fiducial (red dashed = ± 60 MeV/c window, bins $N \geq 20$)

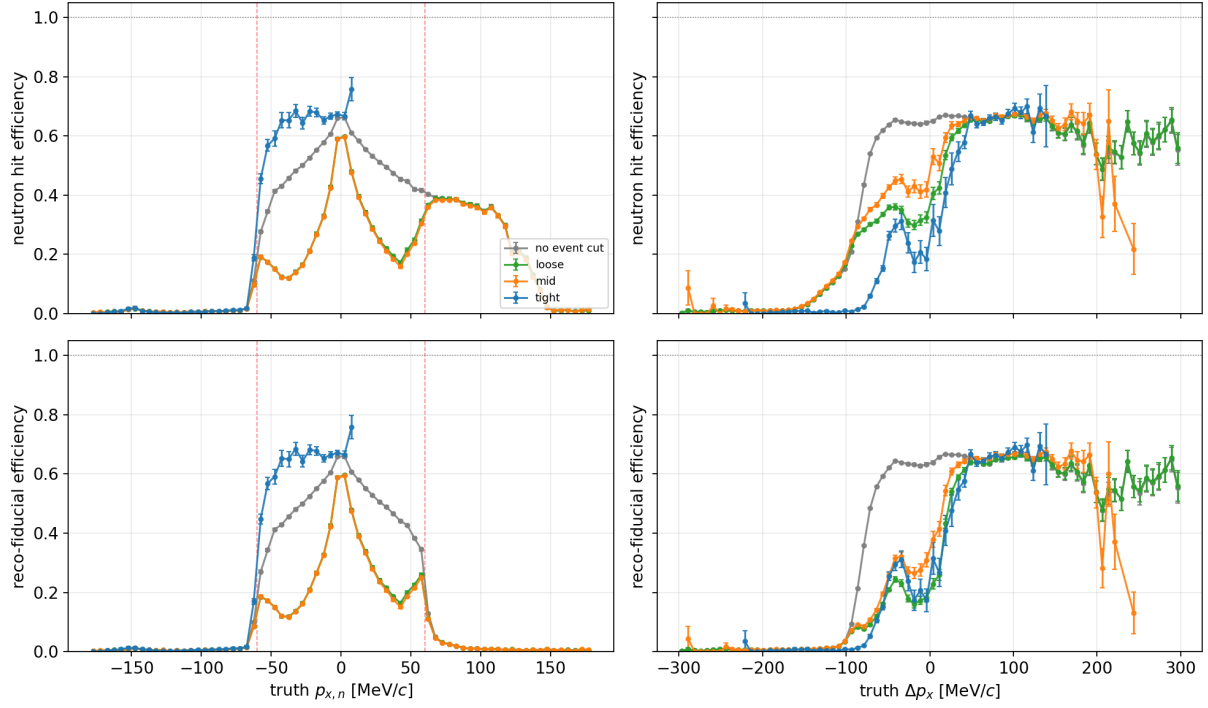


Figure 24: y-pol neutron efficiency across the **full** $p_{x,n}$ range, no px fiducial. Red dashed = ± 60 MeV/c (the px60 window boundary). Top row = neutron hit efficiency, bottom row = reco-fiducial efficiency. Four curves as before (none/loose/mid/tight).

Sn124 zpol: neutron efficiency, NO px fiducial (red dashed = ± 60 MeV/c window, bins $N \geq 20$)

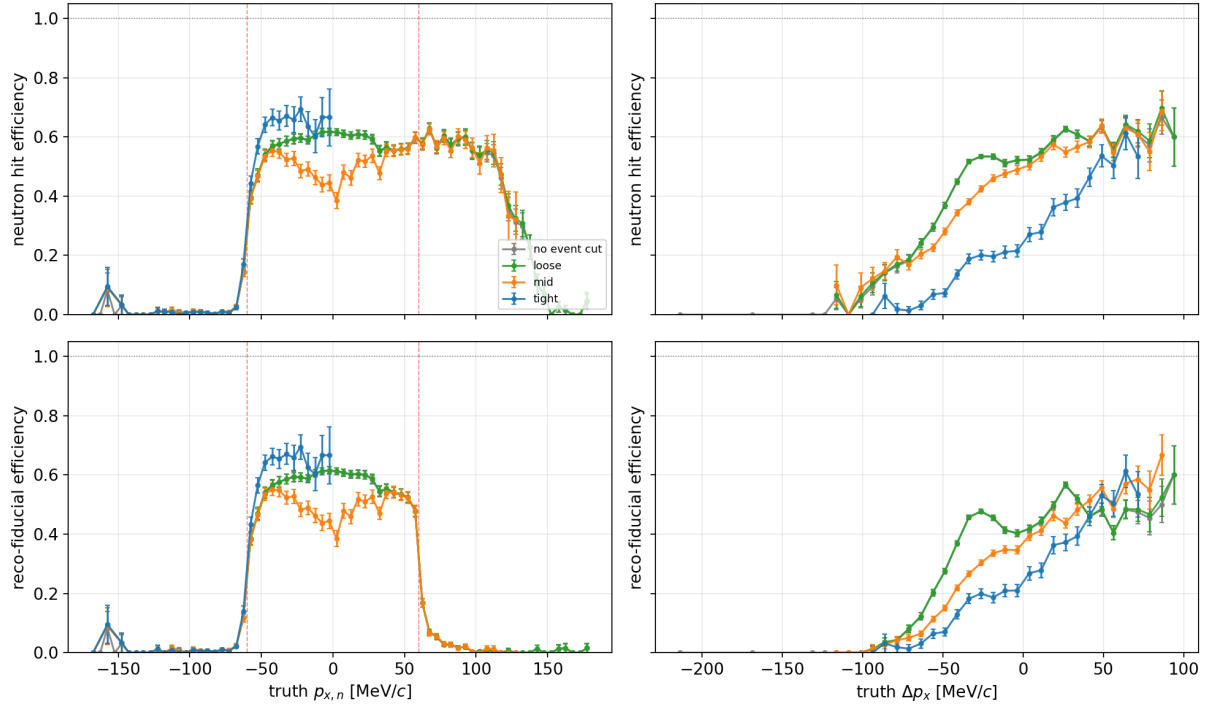


Figure 25: z-pol: the corresponding full-range efficiency curves.

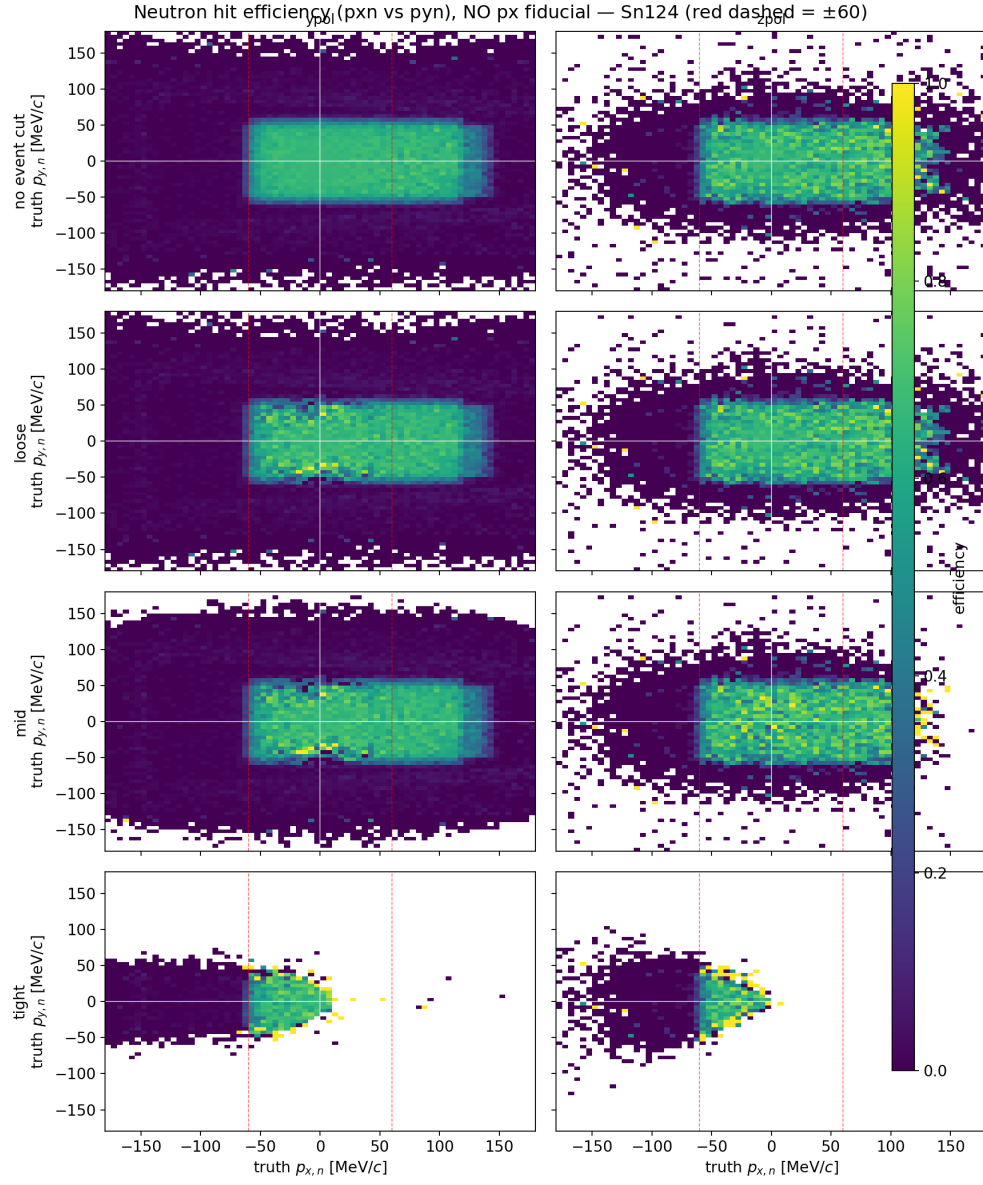


Figure 26: 2D neutron hit efficiency without px fiducial ($p_{x,n}$ vs $p_{y,n}$), one row per event-quality cut. Red dashed = ± 60 MeV/c.

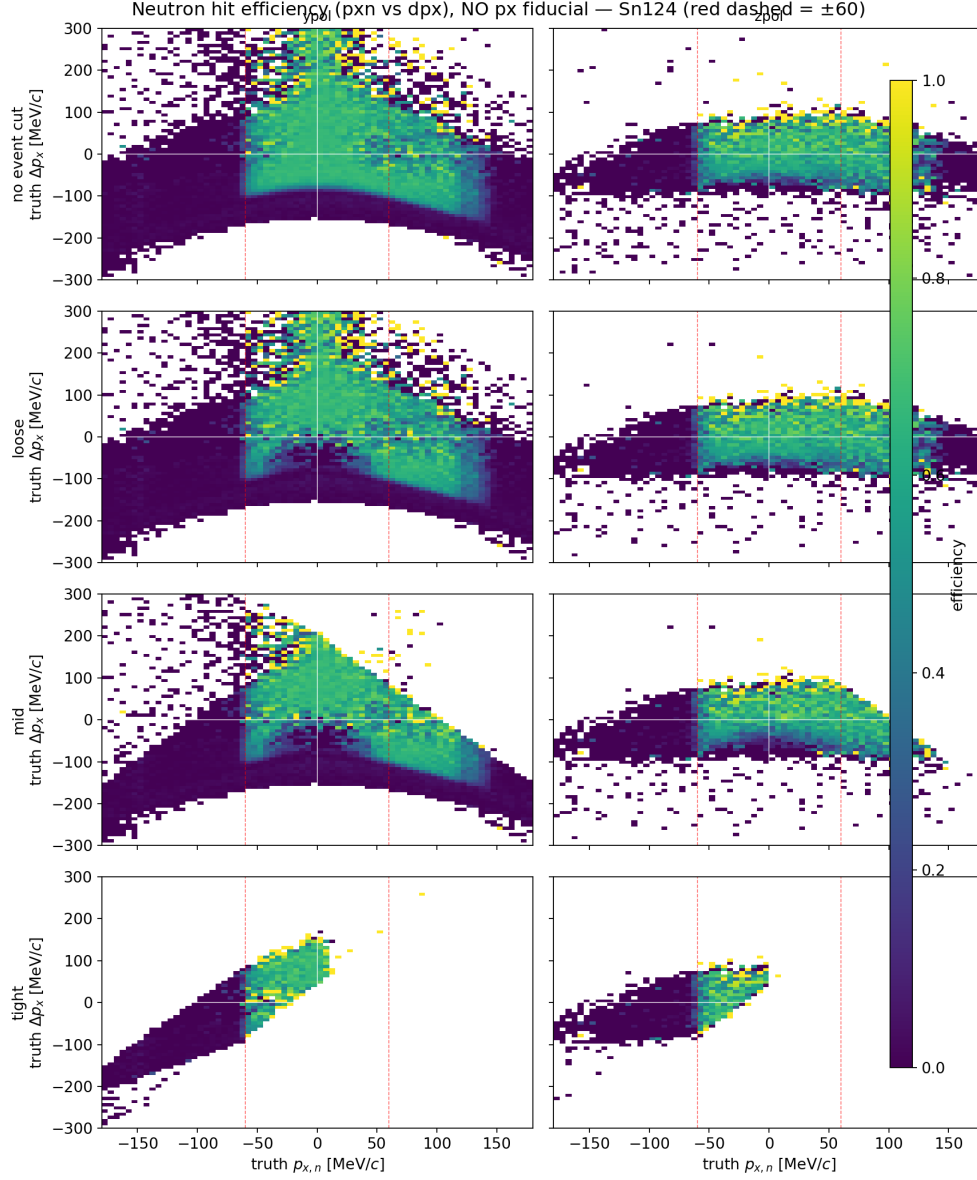


Figure 27: Same, $p_{x,n}$ vs Δp_x .

These figures answer the core question: *why is the px60 cut needed*. With the px fiducial removed, the full neutron-efficiency landscape is visible — NEBULA efficiency is high (~ 0.6 – 0.7) on the positive- $p_{x,n}$ side but drops sharply for negative $p_{x,n}$ with $|p_{x,n}| > 60$ (y-pol). This is exactly why px100 keeps a negative branch that NEBULA can barely see, distorting R . The px60 cut confines the analysis to the core region where both sides have efficiency > 0.5 , keeping the sign-dependent efficiency imbalance within a modelable range.

7 Technical Conventions: Coordinates, p_x , and Reaction Plane

The lab frame is the Geant4/SAMURAI world frame used by the NEBULA-Plus macro: $+Z$ points downstream along the nominal beam direction, $+Y$ is vertical, and $+X$ is the horizontal dispersive direction shown in the event-display projections below. The target/input frame is the frame of the generated or reconstructed target momentum. Unless a label explicitly says “lab”, p_x in this report means p_x^{tgt} , i.e. the target-frame component used for the $p_{x,n}$ fiducial cuts and response plots.

The target angle is $\theta_{\text{tgt}} = 3^\circ$. In the Geant4 primary generator, a particle is started from the target center and the input target-frame direction is rotated by $-\theta_{\text{tgt}}$ about lab Y :

$$\mathbf{p}^{\text{lab}} = R_y(-\theta_{\text{tgt}}) \mathbf{p}^{\text{tgt}}.$$

Therefore positive target-frame p_x means an initial component toward lab $+X$ before the common target-angle rotation. For the simple $p_y = 0$ sign checks,

$$p_x^{\text{lab}} = p_x^{\text{tgt}} \cos \theta_{\text{tgt}} - p_z^{\text{tgt}} \sin \theta_{\text{tgt}},$$

using the same rotation convention as the simulation.

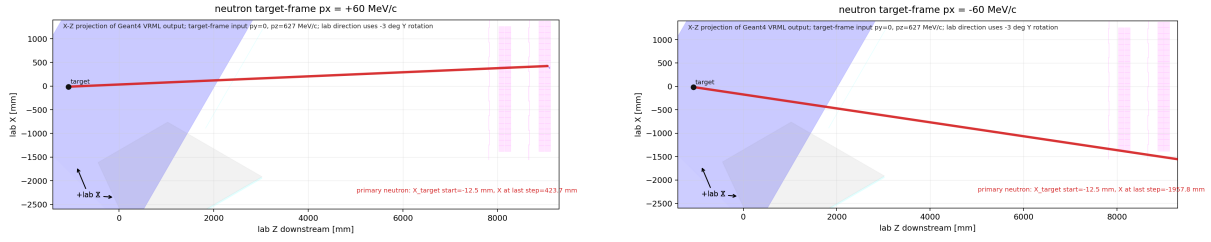


Figure 28: Geant4 VRML-to-PNG projections for target-frame neutron $p_x = \pm 60$ MeV/ c , with $p_y = 0$ and $p_z = 627$ MeV/ c . The horizontal axis is lab Z downstream; the vertical axis is lab X . Positive target-frame p_x trends toward lab $+X$ after the -3° target rotation, while negative target-frame p_x trends toward lab $-X$.

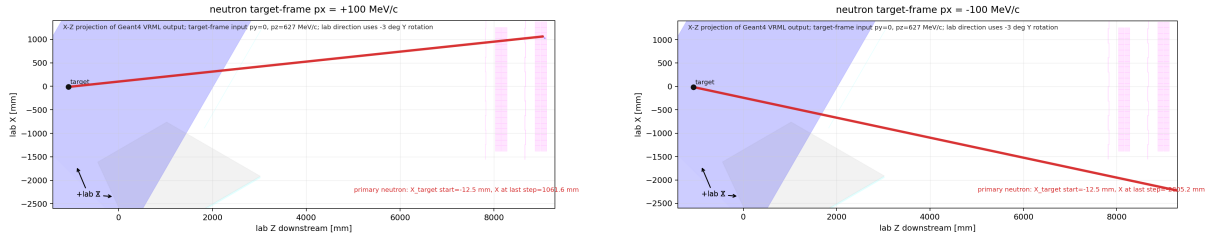


Figure 29: The same Geant4 sign check for target-frame neutron $p_x = \pm 100$ MeV/ c . The larger $|p_x|$ gives the expected larger lab- X deflection, validating the sign convention used in the reconstructed $p_{x,n}$ plots.

Two reaction-plane azimuths must be kept separate. The generator/model angle b_ϕ is the reaction-plane azimuth carried by the QMD/event metadata; in the y-pol raw samples it follows the recorded `rpphi` modulo 2π , while randomized samples carry the randomized plane angle in the event record. This is a truth/model-plane label and is useful for generator checks.

The current folded observable instead uses the event-by-event kinematic plane. Let

$$\mathbf{k}_{\text{out}} = \mathbf{p}_p + \mathbf{p}_n, \quad \mathbf{k}_{\text{out},T} = (k_{\text{out},x}, k_{\text{out},y}).$$

Because $\mathbf{k}_{\text{in}}$ defines the target-frame beam axis, the azimuth of the $(\mathbf{k}_{\text{in}}, \mathbf{k}_{\text{out}})$ reaction plane is

$$\phi_k = \text{atan2}(k_{\text{out},y}, k_{\text{out},x}) = \text{atan2}(p_{y,p} + p_{y,n}, p_{x,p} + p_{x,n}).$$

If one defines a momentum-transfer vector as $\mathbf{q} = \mathbf{k}_{\text{in}} - \mathbf{k}_{\text{out}}$ instead, the transverse azimuth is shifted by π ; this convention should not be mixed with the current R definition.

The rotation used for Δp_x is

$$\begin{pmatrix} p_x^{\text{rot}} \\ p_y^{\text{rot}} \end{pmatrix} = R_z(-\phi_k) \begin{pmatrix} p_x \\ p_y \end{pmatrix} = \begin{pmatrix} p_x \cos \phi_k + p_y \sin \phi_k \\ -p_x \sin \phi_k + p_y \cos \phi_k \end{pmatrix}.$$

The observable is then

$$\Delta p_x = p_{x,p}^{\text{rot}} - p_{x,n}^{\text{rot}}, \quad R = \frac{N(\Delta p_x > 0)}{N(\Delta p_x < 0)}.$$

8 Remaining Work

The current report should not overstate the following points:

- $\eta_{\text{coin}} = 4.65\%$ is still a manuscript planning anchor. The local selected-breakup validation reproduces an acceptance of about 37.9%. These are different definitions and must be kept separate in the proposal.
- Target-specific Sn112 and Sn124 breakup cross sections still need a consistent ImQMD shell-integration extraction. They should not be obtained by simply rescaling the Sn124 anchor.
- Real-data neutron efficiency calibration, timing offsets, thresholds, cross-talk, fake neutrons, and accidental backgrounds are not yet included.
- The detector-response systematic uncertainty has not yet been quantified with toy closure or pseudo-data closure tests.
- The current diagnostic report uses a mixture of simulation truth-defined event cuts and a reconstructed visible-window definition. The final experimental analysis should express all data-side cuts in reconstructed quantities, with simulation truth used only for validation and interpretation.

9 Recommended Proposal Wording

The defensible statement is:

For Sn124, after NEBULA-Plus detector response, PDC NN proton reconstruction, and neutron reconstruction, the folded R observable with the tight event selection and $|p_{x,n}| < 60$ MeV/ c retains a strong γ dependence. Under the current 15 mm/1.2 g Sn target option, 1.0×10^7 pps beam intensity, and 16 h planning assumption, the expected statistics exceed the 3σ adjacent- γ separation requirement by about two orders of magnitude for y-pol ($\sim 99\times$) and about four orders of magnitude for z-pol ($\sim 1.4 \times 10^4\times$). The feasibility of constraining γ is therefore controlled mainly by detector-folding systematics, not by raw statistics.

The following statement should not be used:

The reconstructed R is already equal to the truth-level R .

The more accurate statement is:

The experiment should compare folded data with folded model predictions. The truth-level R should be used as a physics interpretation and closure-test reference.